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BLE-Based Indoor Positioning System: A Low-Cost Alternative to GPS for Indoor Navigation

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Abstract. The Global Positioning System (GPS) uses signals from satellites to specialised devices to detect an entity's location on Earth's surface. However, GPS is ineffective for indoor navigation due to high power consumption and signal obstruction by walls. This study proposes a low-cost, energy-efficient alternative using Bluetooth 4.0-enabled Bluetooth Low Energy (BLE) beacons to enable Indoor Positioning and guide users in confined spaces, such as libraries. Tiny, low-energy BLE beacons transmit Advertising Packets at regular intervals. In contrast, Anchor Nodes send and receive signals to construct a Received Signal Strength Indication (RSSI)-based signal-to-distance model, enabling triangulation-based localisation. The system monitors behaviour and environmental interaction through RSSI analysis and user interaction with Beacons. Proven in a library setting, the BLE-based system proved extremely accurate in guiding users to destinations, bridging GPS shortcomings at minimal energy expenditure and cost.

Keywords. Bluetooth 4.0, Bluetooth Low Energy (BLE), Indoor Positioning, Received Signal Strength Indication (RSSI), Anchor Node, Beacon, Advertising Packets, and Global Positioning System (GPS).

1. Introduction

Users worldwide can obtain exact location and timing information through the ground-breaking satellite-based Global Positioning System (GPS). GPS receivers can precisely locate themselves anywhere on or near the Earth, as long as they have a clear view of the sky, thanks to a constellation of 24 orbiting satellites that continuously send signals [1]. GPS provides unmatched navigational capabilities to military, civilian, and commercial customers. GPS guides and keeps track of users, whether they are soldiers navigating the battlefield, hikers discovering wilderness paths, or delivery drivers planning a route. Nonetheless, keeping an active link with four or more satellites is necessary for accurate positioning. With just three, accuracy is compromised.

GPS is unsuitable for interior monitoring because it depends on satellite signals, which are difficult to receive inside enclosed structures [2]. GPS reception indoors is reduced or obstructed because building materials are difficult for signals to pass through. Multipath interference is

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introduced by walls and ceilings, which results in signal distortion and inaccurate indoor location determination. GPS positioning accuracy is further compromised by structures' poor sight of the satellite constellation. Furthermore, it is not feasible for real-time indoor monitoring applications due to the GPS receivers' initialisation time and ongoing power consumption. In these situations, alternative indoor positioning technologies—such as beacons, WiFi positioning, and RFID—offer more practical options, as they can provide precise location data without being constrained by GPS's Outdoor-focused architecture [3].

Bluetooth Low Energy (BLE) beacons, commonly referred to as beacons, are small radio transmitters powered by batteries that broadcast Bluetooth signals [4,5]. These signals may be scanned for and displayed on Bluetooth-enabled devices. Devices close to the emission region can take up the signal, which helps us determine whether the emitter and receiver are near one another. The Received Signal Strength Indicator (RSSI), a feature of Bluetooth location services, calculates the separation between the emitter and the receiver. Bluetooth Low Energy (BLE) uses substantially less power since it delivers data over a smaller area and at shorter distances [6]. The gadget recognises the beacon's signal and may roughly determine the user's indoor location by calculating the distance to the beacon. We can decide if the emitter and receiver are close to one another by using devices that can detect the signals from the emission region. The distance between the emitter and the receiver is determined using Bluetooth location services' Received Signal Strength Indicator (RSSI).

Two well-liked Bluetooth Low Energy (BLE) beacon protocols are iBeacon and Eddystone. BLE beacons send signals to neighbouring Bluetooth-enabled devices to provide location-based services and proximity marketing. Contextual awareness, indoor placement, and mobile device interaction are made possible by iBeacon and Eddystone. The unique IDs of the beacons, which help identify the beacon closest to the user, are frequently provided to iBeacon and Eddystone. They also send additional identifying information and power. When a beacon is deployed, its physical coordinates for a certain interior setting are assigned to an external database. A device's approximate position can be ascertained by comparing its RSSI, beacon coordinates, and signal values [7]. RSSI is a measurement of beacon signal strength. When the objects are closer together, the value increases, and it decreases when they are farther apart. Around the building, beacons and receivers are strategically positioned to ensure that every area is covered. Thanks to this technology, people using Bluetooth-enabled smart phones will be able to locate one another, which will also send pertinent information to their cell phones based on their whereabouts as they approach designated places.

Beacons are essential for indoor monitoring applications because they provide accurate location-based services and enhance the user experience in confined areas [8]. Beacons enable tailored marketing in retail settings by alerting customers to deals or promotions as they approach specific aisles or products. Beacons are used in museums and exhibitions to transmit interactive content that provides visitors with pertinent information about the exhibits based on their proximity. Beacons can be utilised in the healthcare industry to track the movement of staff members or medical equipment within a facility, thereby enhancing operational effectiveness. Furthermore, beacons make indoor navigation and wayfinding easier in big corporate buildings, directing staff members and guests to their destinations with ease. Beacons are useful in the education sector for monitoring student attendance, maintaining security, and providing location-specific data. Beacons are essential instruments in indoor monitoring because they provide context-aware interactions, real-time tracking, and customised services, all of which help optimise indoor settings.

This study proposes a distributed indoor locating system using Bluetooth Low Energy (BLE) to precisely locate users in a limited space and direct them to the area of interest. It is required of a library customer to select a book that picks their interest. Once the target has been set using the distance computed from the signal strength collected from the beacon and the user, the user is identified and guided to the destination. The digital indoor environment will preserve the item's point of attention. Advancements in Bluetooth technology have made it feasible to provide low-cost indoor positioning and location services that are more accurate, dependable, and precise.

1.1 Related Works

Initially, Indoor Localisation for IoT Applications was proposed by P. S. Varma and V. Anand [9], who mentioned that implementing IoT applications in smart buildings has become considerably more straightforward due to the widespread use of smartphones and other wireless communication capabilities. The limitations of the Global Navigation Satellite System (GNSS) in indoor areas are another reason why indoor positioning systems are gaining popularity, particularly as more governments launch innovative building and city initiatives. The proposal offers a manual site investigation that can be utilised to quickly collect RSSI (Received Signal Strength Indicator) values from WiFi access points; this gathered data can be passed as input to train the prediction models properly. The proposal also offers some insights into the data gathered.

Furthermore, Pascale, F., Adinolfi, E. A., Avagliano, M., Giannella, V., & Salas, A. [10] introduced a Low-Energy IoT Application for Indoor Localisation using Beacons. This study aims to evaluate the most popular techniques, tools, and IoT applications for indoor localisation, while examining the most important existing systems and how these solutions are applied in real-world situations. Then, a novel technique for detecting people flow in an interior setting. Based on Beacon Bluetooth Low Energy (BLE) technology, this work examines the potential for utilising our system in various applications.

F. S. Daníş, A. T. Cemgíl, and C. Ersoy [11] presented a paper on Indoor Positioning and Tracking with Bluetooth Low Energy Beacons using an Adaptive Sequential Monte Carlo Filter. The model treats Bluetooth low-energy (BLE) transmitters as a hidden Markov model with three layers, enabling joint state and parameter estimation. Monte Carlo sampling techniques are used after Bayesian approximation. When used in a test environment with several BLE sensors, the tracking only uses the recorded signals' received signal strength indicator (RSSI). The monitored BLE transmitter is assumed to be silent regarding any further movement. The transition density is essentially a diffusion process in which the probability measurements are dispersed into adjacent space.

Ho, Y. H., & Chan, H. C [12] proposed a Bluetooth Low Energy decentralised adaptive indoor positioning protocol, which provides a BLE-based decentralised mechanism that doesn't require any prior training for use. The anchor nodes can do training autonomously on demand. The anchor nodes simultaneously broadcast and examine for signals sent by other nodes. Each anchor exchanges signal strength measurements and location data. The signal-to-distance reference list is created during this process to determine the physical distance accurately.

Surian, D., Kim, V., Menon, R., Dunn, A. G., Sintchenko, V., & Coiera, E [13] created a new technique based on the received number of signals indicator (RNSI) with BLE beacons and contrasted with a conventional RSSI-based method for tracking a moving user in indoor environments. Some people struggle to navigate indoor spaces. The attention or personalised guides that help guide in these

situations are almost non-existent in some cases, causing the user to give up their search for a solution to this problem.

B. P. Guamán and J. Cordero's [14] paper presents a solution to the problem mentioned earlier by designing and implementing an indoor positioning system based on Beacon technology. The system is divided into two parts: a web application for information management and a mobile app for interacting with beacons to direct users to their destinations or display notifications about their current location. Kaluža, Marin & Vukelic, and Bernard [15] analysed various indoor positioning systems and IPS concepts. It shows technologies that can be employed to implement the IPS system and evaluates the categories that influence implementation. The paper demonstrates that BLE performed best in several individual categories, while none of the technologies met all the observed characteristics.

E. Essa, B. A. Abdullah, and A. Wahba [16], stated that for indoor positioning systems (IPS), BLE is a useful solution that provides acceptable accuracy and affordable deployment. The Received Signal Strength Indicator (RSSI) is used to approximate the distance between various devices, and the RSSI is applied with a Kalman Filter to enhance the correctness of the measured distance. The trilateration method is used to estimate device positions depending on previous results. The experiment results show an average positioning accuracy ranging from 0.76 to 0.39 m.

The overview of various research contributions to indoor localisation for IoT applications reveals a consensus on the challenges and opportunities associated with leveraging emerging technologies for precise indoor positioning. Collectively, these studies emphasise the practicality and effectiveness of BLE technology for indoor localisation, showcasing diverse approaches to address the challenges associated with precise indoor localisation in the era of IoT.

2. Architecture Diagram

Figure 1 elaborates on the components required and the flow involved in implementing a distributed indoor location system based on Bluetooth Low Energy (BLE) from the library.

2.1 Central Components

This study's central components consist of three key elements, viz., the user, the book, and the Database. Users are represented by a stick figure holding a book, likely the library patron searching for information. The book symbolises the physical library collection, containing the desired information. The Database is illustrated as a computer terminal, representing the library's digital collection and potentially other relevant online resources.

2.2 Data Flow

When a beacon app-level program exists, it is either built into the mobile application used or it makes use of a separate device. Beacons are small transmitters strategically placed within the library building. They emit Bluetooth or radio signals, which create invisible waves to interface with the user device. The strength of these signals is known as RSSI-values or Received Signal Strength.

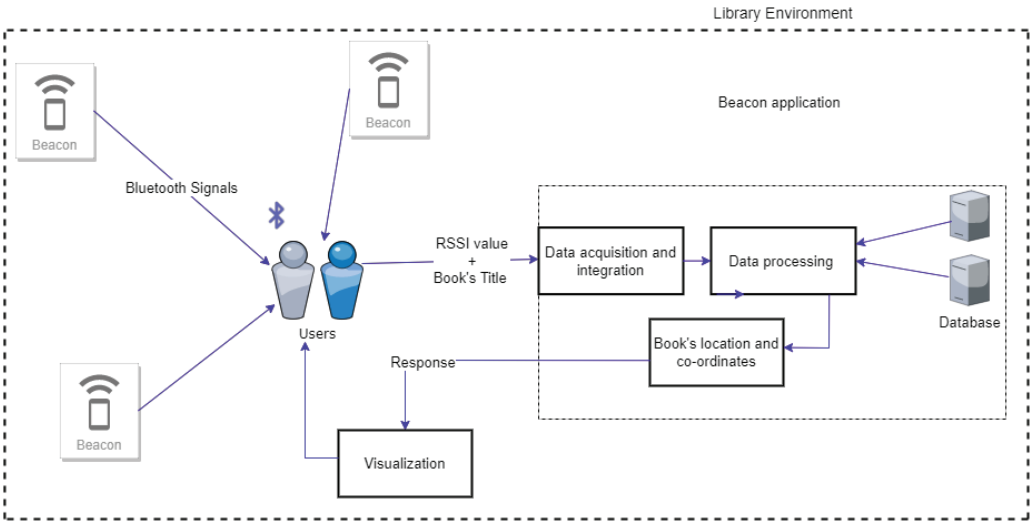


Figure 1. System Architecture of the proposed system

Indicator values serve as a basis for estimating the user's position within the building. When the system acquires RSSI values from several beacons and merges them, it can track the user's position in the building with great accuracy. These location data will enable the application to access the library database and retrieve useful information, such as the preferred title and precise shelf coordinates of a desired book. This information, including the specific shelf and location of the book, will then be displayed on the user's gadget or an information screen in the library. The user will then either use the information to move towards the book or view more facts about it.

2.3 Overall Process

The user activates the beacon application on their device. Beacons transmit Bluetooth signals, and the user's device receives them with varying strengths depending on proximity. The app processes the signal strengths and calculates the user's location within the library. The user's location is used to query the library's Database for relevant information, such as the title and location of the book they might be searching for. The book's location and potentially other details are displayed on the user's device or a library information screen. The user interacts with the provided information, finds the book, accesses additional details, or takes other actions. As a means of communication between beacons, packets will be continuously transmitted. RSSI values will be used to estimate the user's position. The map to the chosen book location from the Database will be displayed to the user from the user's current position. Figure 2 illustrates the system's flow.

3. Methodology

The proposed system reduces the need for manual book searching in a library. Users cannot manually search a book through a large library's shelves. It's a time-consuming process that may lose the user's interest. It is assumed that the precise location of books is stored in a database. This system includes a beacon placed at each library entrance. The user's position is estimated based on the RSSI from these beacons. Beacons were used because they do not require any infrastructure, consume little power, and can run at a minimal cost for extended periods. The user can choose a book from the available list of books in the UI. The algorithm then runs in the background, displaying the path from the computed

user's current location and navigating the user to the desired book location. A star algorithm is used because it finds the optimal path from the user to the book. The system's operation is divided into three stages: identifying the user's position, identifying the book's position, and finding the path from the user to the book.

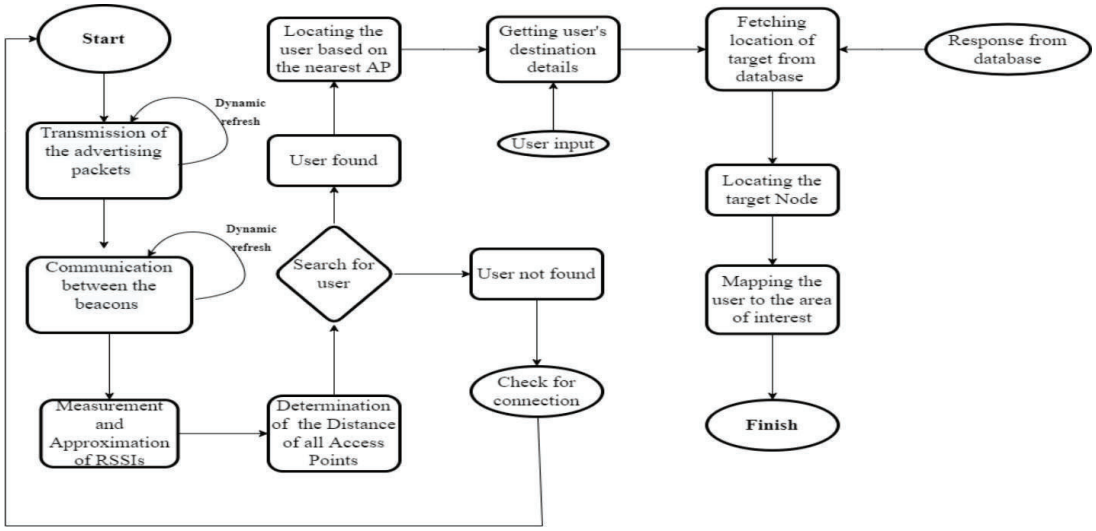


Figure 2. System Workflow

Step 1: Finding User Position

The user enters the library with Bluetooth enabled and the developed application installed on his device. The beacons were placed in the library environment at various locations (i.e., different entrances), as shown in Figure 3, and the beacon specifications are shown in Figure.4.

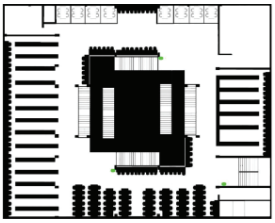


Figure 3. Beacon positions in the library

Brand	Blue Charm Beacons
Manufacturer	Blue Charm Beacons
Item Height	6 Millimeters
Item Width	36 Millimeters
Product Dimensions	3.6 x 3.6 x 0.6 cm; 10 Grams
Batteries	1 Lithium Metal batteries required. (included)
Item model number	BCB-iBeacon-037
Wireless Type	Bluetooth
Are Batteries Included	Yes
Lithium Battery Energy Content	0.67 Watt Hours
Lithium Battery Voltage	3 Volts
Lithium Battery Weight	0.11 Grams
Number Of Lithium Ion Cells	1
Number of Lithium Metal Cells	1
Manufacturer	Blue Charm Beacons
Country of Origin	USA
Item Weight	10 g

Figure 4. Beacon Specifications

The anchor nodes send advertising packets to all other anchor nodes within their range. While broadcasting, the beacons observe for signals transmitted by other beacons. This cooperative method of communication allows each beacon to exchange location information and signal strength measurements. A unique identification number identifies the beacons. The anchor nodes now start finding the RSSI of the user's device. The developed application will collect RSSI values from all beacons every 10 seconds. These RSSIs are stored in a CSV file, and the mode of these values is used to estimate the exact RSSI of the user from different beacons. The Received Signal Strength Indicator (RSSI) values estimate the distance.

$$\mathrm{RSSI@d} = \mathrm{RSSI@dr} - 10\mathrm{n}\log_{10}\left(\frac{\mathrm{d}}{\mathrm{dr}}\right) \quad (1)$$

Eq. (1) represents the relationship between RSSI and the distance based on the long-distance path loss model, where: $RSSI@dr$ is the reference RSSI value, $RSSI@d$ is the RSSI value calculated, dr is the reference distance, and d is the current distance.

Step 2: Finding the path from the user location to the desired book

After the user selects a book from the GUI, the "A Star" algorithm uses the start coordinate measured by Step 1 and the end coordinate measured by Step 2 to determine whether a path is available to that book. If a path is available, the algorithm will output an image along with the path.

Initially, the image is read by an algorithm and converted into a binary image. The blacks are immovable regions, and the whites are movable pixels. Every pixel will have 0 or from the start node, the algorithm checks if it is movable in all four directions. If so, the positions are added to a list. The algorithm now iterates through each value in the list, checking if it has been visited before and if it is within range. If it has not been visited and is within range, the algorithm calculates the total cost to reach it, finds the minimum-cost neighbour, and traverses to that node. Then, the process is repeated until the end coordinate is reached. The total cost is the sum of the Euclidean distance and the heuristic function.

Eq. (2) represents the relationship used in the A* search algorithm, where: child.f is the total cost to traverse the chosen neighbour, child.g is the cost to traverse one neighbour, and child.h is the Euclidean distance between the chosen neighbour and the end coordinates.

$$\mathrm{child.f} = \mathrm{child.g} + \mathrm{child.h} \quad (2)$$

The path returned by the algorithm is then displayed to the user as an image as shown in Figure 5. Table 1 shows the comparison between the proposed system and the existing systems.

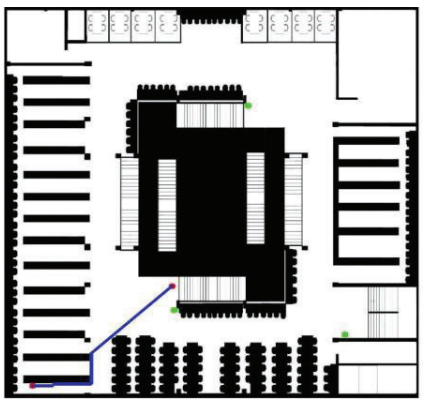


Figure 5. Sample result

4. Results and Discussions

The User interface to enter the preferred book is shown in Figure 6. From Figure 7, the user is near the 1st staircase, so the beacon-1 RSSI will be higher compared to the other two beacons. Hence, it is determined that the user is near Beacon-1, and the user's start location is identified. Now that the user has selected the book, the program will automatically determine its end location from the Database. A star algorithm runs behind, finds the shortest path, and displays it to the user. The red dots represent the start and end points. The green-colored dots are beacons. The blue line shows the path to be followed by the user.

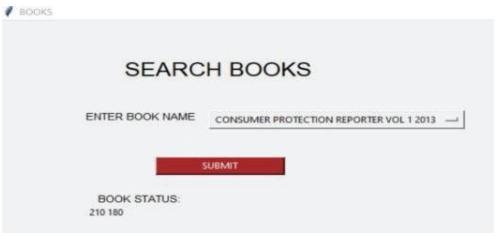


Figure 6. Graphical User Interface

In Figure 8, the user is near the 3rd staircase, so the beacon-3 RSSI will be higher compared to the other two beacons. Hence, it is identified that the user is near beacon-3, and the user's start location is determined. Now that the user has selected the book, the program will automatically determine its end location from the Database. A star algorithm runs behind, finds the shortest path, and displays it to the user. The red dots represent the start and end points. The green-colored dots are beacons. The blue line shows the path to be followed by the user.

Table 1. Comparison of the Proposed Architecture with Existing Systems

Dimension	**Proposed architecture**	**[10]**	**[11]**	**[12]**	**[13]**	**[14]**	**[16]**
Scope	Full pipeline + visualization + DB	Review + concept	Algorithm only	Protocol + anchors	Metric study	System demo	Algorithm + accuracy report
Algorithm choice	Pluggable/fusion (RSSI/KF, RNSI, SMC)	Not fixed	SMC only	Decentralized learning	RNSI focus	Basic nav	RSSI+KF
Deployment effort	Low (drop beacons; phone as scanner)	Conceptual	Higher (compute/ tuning)	Medium (anchor coordination)	Low	Medium (two apps)	Low–Medium (calibration)
Adaptation /calibration	Data-driven via DB, zone rules	Limited	Needs tuning	Self-train anchors idea can be adopted	Limited	Manual	Manual calibration
UX /Visualization	Built-in (real-time guidance)	No	No	No	No	Yes	No
Maintainability	High (logs, retraining, A/B in DB)	N/A	Medium	Medium	Medium	Medium	Medium
Accuracy potential	Equal/Better by choosing per-zone method & fusing	Not quantified	Strong in noise, higher compute	Improves over time	Stable but coarse	Placement-dependent	Reported 0.39–0.76 m

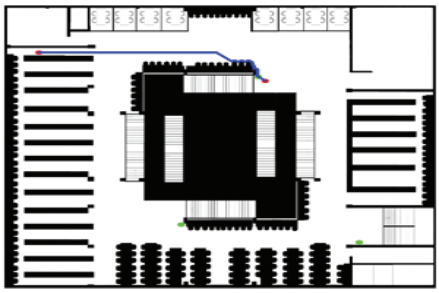


Figure 7. The shortest path from entry point1

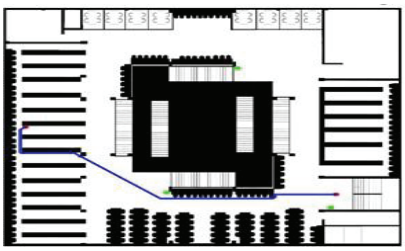


Figure 8. Shortest path from entry point2

In Figure 9, the user is near the 2nd staircase, so the beacon-2 RSSI will be higher compared to the other two beacons. Hence, it is identified that the user is near beacon-2, and the user's start location is determined. Now that the user has selected the book, the program will automatically determine its end location from the Database. A star algorithm runs behind, finds the shortest path, and displays it to the user. The red dots represent the start and end points. The green-colored dots are beacons. The blue line shows the path to be followed by the user.

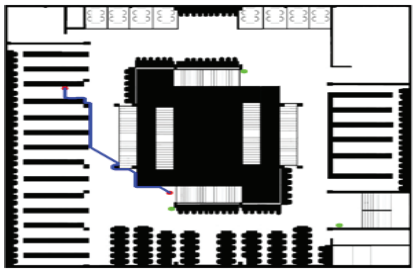


Figure 9. Shortest path from entry point 3

5. Conclusion

To track and guide users within a library to their chosen book, BLE (Bluetooth Low Energy) technology is employed due to its effective performance for indoor location purposes. Beacons, integral to this system, outperform WiFi in terms of accuracy, battery life, and processing speed. A CAD model tailored for the library environment incorporates numerous strategically placed beacons. Upon entering the library, users are prompted to run a dedicated beacon application to collect Received Signal Strength Indicator (RSSI) data from nearby beacons, enabling precise location determination. The predefined dataset is used to identify the book's location, and image processing is then used to map the user's final path to the book. The following developments are planned for the

project: By including RFID tags in each book and fixing a global RFID, it is expected that when a user selects a book, the RFID tag will activate and transmit a response signal to the global RFID, letting it know where it is. As a result, the user can put a book anywhere in the library and find it anywhere else in the building. It is intended to develop this technology into a three-dimensional architecture.

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